

LAURENTIAN FAN BUOY 44141, Canada

WMO: 996260

Lat: 43.000N

Lon: 58.000W

Elev: 10

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.5	28.1	N/A	N/A	N/A	N/A	N/A	N/A	44.1	42.8	41.4	40.2	20.9	300	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	76.3	N/A	75.1	N/A	73.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 39.0	34.8	31.2	DB	23.9	76.8	4.0	2.7	21.0	78.7	18.7	80.3	16.4	81.8	13.5	83.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	43.2	40.9	41.1	44.0	49.8	57.9	66.6	69.3	65.8	59.4	53.3	47.3		
	DBStd	11.52	6.81	7.78	7.25	5.39	5.34	5.33	4.25	3.22	4.08	5.52	6.06	6.95		
	HDD50	1229	226	269	291	196	70	3	0	0	0	3	35	136		
	HDD65	4580	677	674	741	631	473	223	30	4	39	186	352	550		
	CDD50	2428	15	16	14	14	62	239	515	598	473	296	134	52		
	CDD65	303	0	0	0	0	0	10	80	137	61	14	1	0		
	CDH74	218	0	0	0	0	0	1	67	124	26	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	14.8	18.1	17.9	16.8	14.2	11.9	11.2	11.1	11.1	13.4	16.6	16.6	18.4		
(15) Precipitation	PrecAvg	62.3	6.1	4.9	4.5	3.5	3.7	4.5	4.9	5.5	4.5	5.9	6.8	7.4		
	PrecMax	79.9	9.7	7.4	7.0	7.1	6.7	9.3	10.2	11.7	13.7	12.1	10.8	11.8		
	PrecMin	47.8	2.4	2.8	2.3	0.9	0.7	1.5	1.6	1.6	0.9	1.1	3.3	3.5		
	PrecStd	8.2	1.7	1.3	1.2	1.4	1.5	2.1	2.1	2.3	2.8	2.6	2.4	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.1	63.6	64.1	63.5	67.1	73.1	77.2	78.0	76.5	72.6	69.9	64.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	59.8	60.1	60.3	59.4	63.8	70.6	75.9	76.5	74.6	71.1	67.5	62.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	57.2	57.4	56.8	56.0	60.8	68.2	74.5	75.5	73.1	69.8	65.1	60.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	54.3	54.1	53.0	52.2	57.5	66.5	73.2	74.4	71.9	67.8	62.6	58.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	10.1	8.9	6.7	6.1	5.4	4.3	4.1	4.9	6.3	7.8	9.8		
		MCDBR	16.1	15.1	15.9	13.8	10.4	5.6	5.0	5.2	6.1	7.9	11.1	14.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.332	0.351	0.379	0.388	0.406	0.432	0.398	0.378	0.363	0.339	0.327		
	taud		2.421	2.483	2.448	2.394	2.411	2.396	2.374	2.454	2.488	2.504	2.514	2.470		
	Ebn at Noon		250	275	283	282	281	275	266	272	271	260	250	244		
	Edh at Noon		25	27	32	36	36	37	37	34	30	26	23	22		
(44) All-Sky Solar Radiation	RadAvg		371	594	1001	1369	1637	1715	1752	1631	1353	844	507	325		
	RadStd		30	62	90	111	117	150	156	112	108	65	60	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page